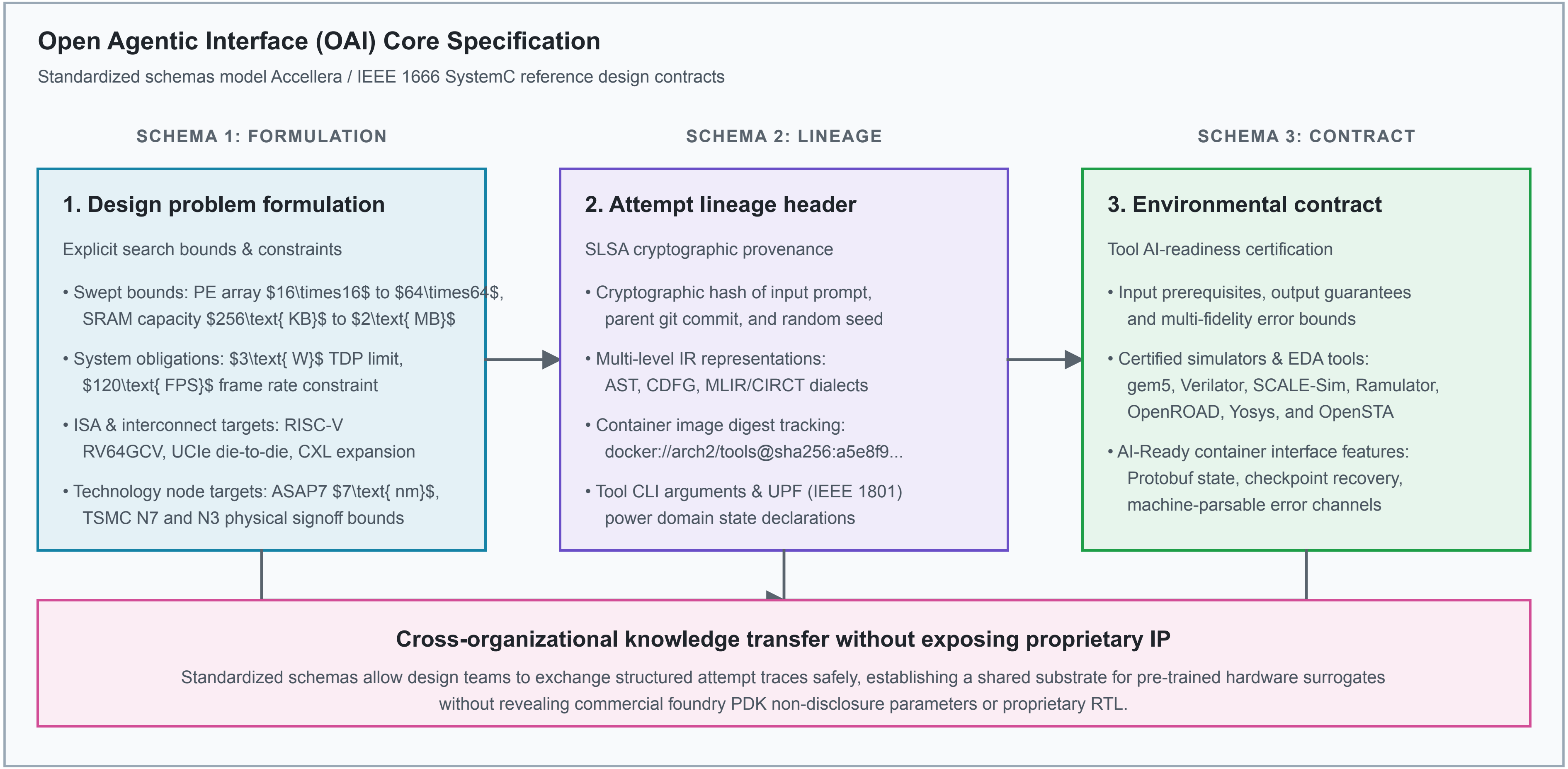
Open Agentic Interface (OAI) Core Specification
Standardized schemas model Accellera / IEEE 1666 SystemC reference design contracts
SCHEMA 1: FORMULATION
SCHEMA 2: LINEAGE
SCHEMA 3: CONTRACT
1. Design problem formulation
Explicit search bounds & constraints
• Swept bounds: PE array 16×16 to 64×64, SRAM capacity 256 KB to 2 MB
• System obligations: 3 W TDP limit, 120 FPS frame rate constraint
• ISA & interconnect targets: RISC-V RV64GCV, UCIe die-to-die, CXL expansion
• Technology node targets: ASAP7 7 nm, TSMC N7 and N3 physical signoff bounds
2. Attempt lineage header
SLSA cryptographic provenance
• Cryptographic hash of input prompt, parent git commit, and random seed
• Multi-level IR representations: AST, CDFG, MLIR/CIRCT dialects
• Container image digest tracking: docker://arch2/tools@sha256:a5e8f9...
• Tool CLI arguments & UPF (IEEE 1801) power domain state declarations
3. Environmental contract
Tool AI-readiness certification
• Input prerequisites, output guarantees and multi-fidelity error bounds
• Certified simulators & EDA tools: gem5, Verilator, SCALE-Sim, Ramulator, OpenROAD, Yosys, and OpenSTA
• AI-Ready container interface features: Protobuf state, checkpoint recovery, machine-parsable error channels
Cross-organizational knowledge transfer without exposing proprietary IP
Standardized schemas allow design teams to exchange structured attempt traces safely, establishing a shared substrate for pre-trained hardware surrogates without revealing commercial foundry PDK non-disclosure parameters or proprietary RTL.